

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

Which of the following expressions has a factor of $x + 2b$, where b is a positive integer constant?

Answer

- A. $3x^2 + 7x + 14b$
- B. $3x^2 + 28x + 14b$
- C. $3x^2 + 42x + 14b$
- D. $3x^2 + 49x + 14b$

Correct Answer: D

Rationale

Choice D is correct. Since each choice has a term of $3x^2$, which can be written as $(3x)(x)$, and each choice has a term of $14b$, which can be written as $(7)(2b)$, the expression that has a factor of $x + 2b$, where b is a positive integer constant, can be represented as $(3x + 7)(x + 2b)$. Using the distributive property of multiplication, this expression is equivalent to $3x(x + 2b) + 7(x + 2b)$, or $3x^2 + 6xb + 7x + 14b$. Combining the x-terms in this expression yields $3x^2 + (7 + 6b)x + 14b$. It follows that the coefficient of the x-term is equal to $7 + 6b$. Thus, from the given choices, $7 + 6b$ must be equal to $7, 28, 42$, or 49 . Therefore, $6b$ must be equal to $0, 21, 35$, or 42 , respectively, and b must be equal to $\frac{0}{6}, \frac{21}{6}, \frac{35}{6}$, or $\frac{42}{6}$, respectively. Of these four values of b , only $\frac{42}{6}$, or 7 , is a positive integer. It follows that $7 + 6b$ must be equal to 49 because this is the only choice for which the value of b is a positive integer constant. Therefore, the expression that has a factor of $x + 2b$ is $3x^2 + 49x + 14b$.

Choice A is incorrect. If this expression has a factor of $x + 2b$, then the value of b is 0 , which isn't positive.

Choice B is incorrect. If this expression has a factor of $x + 2b$, then the value of b is $\frac{21}{6}$, which isn't an integer.

Choice C is incorrect. If this expression has a factor of $x + 2b$, then the value of b is $\frac{35}{6}$, which isn't an integer.